

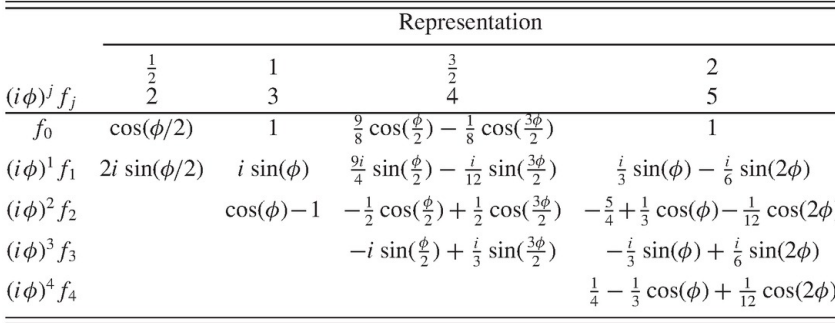
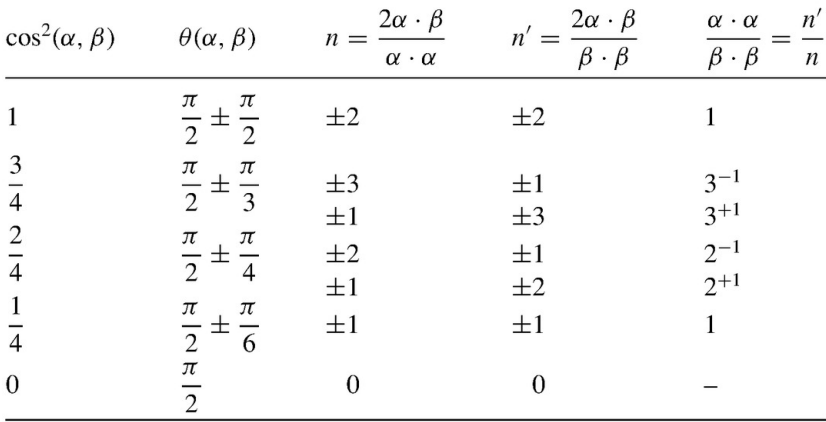
gilmore-lie-groups: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ TAB_001	26	0.877	$\begin{tabular}{t}{ l l l l l }\hline & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{tabular}$	$\begin{array}{c ccc c} & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{array}$	$\begin{array}{c ccc c} & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{array}$
gilmore-lie-groups_ TAB_002	30	0.669	$\begin{tabular}{t}{ l l l l l l }\hline & (I) & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{tabular}$	$\begin{array}{c ccccc c} & I & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{array}$	$\begin{array}{c ccccc c} & I & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{array}$
gilmore-lie-groups_ TAB_003	62	0.837	$\begin{tabular}{t}{ l l }\hline & \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{tabular}$	$\begin{array}{c c } \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{array}$	$\begin{array}{c c } \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{array}$
gilmore-lie-groups_ TAB_004	62	1.000	$\begin{tabular}{t}{ l l l }\hline & \text{Transformation} & \text{F} = d\mathbf{p}/dt & (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{tabular}$	$\begin{array}{c cc } \text{Transformation} & \text{F} = d\mathbf{p}/dt & (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{array}$	$\begin{array}{c cc } \text{Transformation} & \text{F} = d\mathbf{p}/dt & (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{array}$

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gilmore-lie-groups_ TAB_005	86	1.000	$\begin{tabular}{t l l l l l } \hline \multicolumn{2}{ c }{\text{In group}} & \text{Relation} & \multicolumn{3}{ c }{\text{In algebra}} \\ \hline S^t = S^{+1} & \det(S) = +1 & S = e^\Sigma & \text{Tr } \Sigma = 0 & \Sigma^t = +\Sigma & \\ O^t = O^{-1} & \det(O) = +1 & O = e^A & \text{Tr } A = 0 & A^t = -A & \\ \hline \end{tabular}$	$\begin{array}{c cccccc} & X_\eta & X_R & X_L & X_r & X_l & X_\delta \\ \hline X_\eta & 0 & 2X_R & -2X_L & X_r & -X_l & 0 \\ X_R & & 0 & -4X_\eta & 0 & -2X_r & 0 \\ X_L & & & 0 & 2X_l & 0 & 0 \\ X_r & & & & 0 & -X_\delta & 0 \\ X_l & & & & & 0 & 0 \\ X_\delta & & & & & & 0 \end{array}$	
gilmore-lie-groups_ TAB_006	90	1.000	$\begin{tabular}{t l l l l l l } \hline & X_\eta & X_R & X_L & X_r & X_l & X_\delta \\ \hline X_\eta & 0 & 2X_R & -2X_L & X_r & -X_l & 0 \\ X_R & & 0 & -4X_\eta & 0 & -2X_r & 0 \\ X_L & & & 0 & 2X_l & 0 & 0 \\ X_r & & & & 0 & -X_\delta & 0 \\ X_l & & & & & 0 & 0 \\ X_\delta & & & & & & 0 \end{tabular}$	$\begin{array}{c cccccc} & X_\eta & X_R & X_L & X_r & X_l & X_\delta \\ \hline X_\eta & 0 & 2X_R & -2X_L & X_r & -X_l & 0 \\ X_R & & 0 & -4X_\eta & 0 & -2X_r & 0 \\ X_L & & & 0 & 2X_l & 0 & 0 \\ X_r & & & & 0 & -X_\delta & 0 \\ X_l & & & & & 0 & 0 \\ X_\delta & & & & & & 0 \end{array}$	
gilmore-lie-groups_ TAB_007	90	0.600	$\begin{tabular}{t l l l l l l } \hline & \hat{n} + \frac{1}{2}I & a^\dagger a^\dagger & aa & a^\dagger & a & I \\ \hline \hat{n} + \frac{1}{2}I & 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a & 0 \\ a^\dagger a^\dagger & & 0 & -4\left(\hat{n} + \frac{1}{2}I\right) & 0 & -2a^\dagger & 0 \\ aa & & & 0 & 2a & 0 & 0 \\ a^\dagger & & & & 0 & -I & 0 \\ a & & & & & 0 & 0 \\ I & & & & & & 0 \end{tabular}$	$\begin{array}{c cccccc} & \hat{n} + \frac{1}{2}I & a^\dagger a^\dagger & aa & a^\dagger & a & I \\ \hline \hat{n} + \frac{1}{2}I & 0 & 2a^\dagger a^\dagger & -2aa & a^\dagger & -a & 0 \\ a^\dagger a^\dagger & & 0 & -4\left(\hat{n} + \frac{1}{2}I\right) & 0 & -2a^\dagger & 0 \\ aa & & & 0 & 2a & 0 & 0 \\ a^\dagger & & & & 0 & -I & 0 \\ a & & & & & 0 & 0 \\ I & & & & & & 0 \end{array}$	
gilmore-lie-groups_ TAB_008	100	0.817	$\begin{tabular}{t l l l l } \hline & \mu & \sigma & \text{Algebra} & \text{Singular subspace} \\ \hline +1 & +1 & \mathfrak{so}(3,2) & & \\ -1 & +1 & \mathfrak{so}(4,1) & & \\ -1 & -1 & \mathfrak{so}(5) & & \\ +1 & 0 & \text{Poincare} & \text{translations } t_\mu & \\ 0 & 0 & \text{Galilei} & \text{translations } t_\mu, \text{ boosts } \mathbf{b} & \\ \hline \end{tabular}$	$\begin{array}{c cccc} \mu & \sigma & \text{Algebra} & \text{Singular subspace} \\ \hline +1 & +1 & \mathfrak{so}(3,2) & \\ -1 & +1 & \mathfrak{so}(4,1) & \\ -1 & -1 & \mathfrak{so}(5) & \\ +1 & 0 & \text{Poincare} & \text{translations } t_\mu \\ 0 & 0 & \text{Galilei} & \text{translations } t_\mu, \text{ boosts } \mathbf{b} \\ \hline \end{array}$	

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gilmore-lie-groups_ TAB_010	167	0.608	$\begin{array}{l} \begin{array}{ c c c c } \hline \left[H_1, H_2\right] \&= \& 0 \\ \hline \left[\mathbf{H}, E_{\pm 2 \mathbf{e}_1}\right] \&= \& \left(\pm 2 / \sqrt{12}, 0\right) E_{\pm 2 \mathbf{e}_1} \\ \hline \left[\mathbf{H}, E_{\pm 2 \mathbf{e}_2}\right] \&= \& \left(0, \pm 2 / \sqrt{12}\right) E_{\pm 2 \mathbf{e}_2} \\ \hline \left[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}\right] \&= \& \left(\pm / \sqrt{12}, \pm / \sqrt{12}\right) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ \hline \left[E_{+2 \mathbf{e}_1}, E_{-2 \mathbf{e}_1}\right] \&= \& \left(2 / \sqrt{12}\right) H_1 \\ \hline \left[E_{+2 \mathbf{e}_2}, E_{-2 \mathbf{e}_2}\right] \&= \& \left(2 / \sqrt{12}\right) H_2 \\ \hline \left[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}\right] \&= \& \left(1 / \sqrt{12}\right)\left(\pm H_1 \pm H_2\right) \\ \hline \left[E_{+2 \mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}\right] \&= \& *\left(1 / \sqrt{6}\right) E_{\mathbf{e}_1-\mathbf{e}_2} \\ \hline \left[E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}\right] \&= \& \left(1 / \sqrt{6}\right) E_{-2 \mathbf{e}_1} \\ \hline \left[E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2 \mathbf{e}_1}\right] \&= \& \left(1 / \sqrt{6}\right) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ \hline \left[E_{-2 \mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}\right] \&= \& \left(-1 / \sqrt{6}\right) E_{-\mathbf{e}_1-\mathbf{e}_2} \\ \hline \left[E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}\right] \&= \& \left(-1 / \sqrt{6}\right) E_{+2 \mathbf{e}_1} \\ \hline \left[E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2 \mathbf{e}_2}\right] \&= \& \left(-1 / \sqrt{6}\right) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ \hline \left[E_{+2 \mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}\right] \&= \& *\left(1 / \sqrt{6}\right) E_{-\mathbf{e}_1+\mathbf{e}_2} \\ \hline \left[E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}\right] \&= \& \left(1 / \sqrt{6}\right) E_{-2 \mathbf{e}_2} \\ \hline \left[E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2 \mathbf{e}_2}\right] \&= \& \left(1 / \sqrt{6}\right) E_{+\mathbf{e}_1+\mathbf{e}_2} \\ \hline \left[E_{-2 \mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}\right] \&= \& \left(-1 / \sqrt{6}\right) E_{+\mathbf{e}_1-\mathbf{e}_2} \\ \hline \left[E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}\right] \&= \& \left(-1 / \sqrt{6}\right) E_{+2 \mathbf{e}_2} \\ \hline \left[E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2 \mathbf{e}_2}\right] \&= \& \left(-1 / \sqrt{6}\right) E_{-\mathbf{e}_1-\mathbf{e}_2} \\ \hline \end{array} \end{array}$	<table><tr><td>$[H_1, H_2]$</td><td>=</td><td>0</td></tr><tr><td>$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$</td><td>=</td><td>$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$</td></tr><tr><td>$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$</td><td>=</td><td>$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$</td></tr><tr><td>$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm 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E_{-\mathbf{e}_1+\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$	$[E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2}$
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$[E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2}$																																																											

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gilmore-lie-groups_ TAB_011	172	■1.000	<pre>\begin{tabular}[t]{ l l l l l } \hline \multirow{4}[3]{*}{\((i \ \phi)^j \ f_j\)} & \multicolumn{4}{ c }{Representation} \\ \hline & \frac{1}{2} & 1 & \frac{3}{2} & 2 \\ \hline & 2 & 3 & 4 & 5 \\ \hline & \cos(\phi/2) & 1 & \frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2}) & 1 \\ (i \ \phi)^1 f_1 & 2i \sin(\phi/2) & i \sin(\phi) & \frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2}) & \frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi) \\ (i \ \phi)^2 f_2 & & \cos(\phi) - 1 & -\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2}) & -\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi) \\ (i \ \phi)^3 f_3 & & & -i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2}) & -\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi) \\ (i \ \phi)^4 f_4 & & & & \frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi) \\ \hline \end{tabular}</pre>	(not rendered)																																									
gilmore-lie-groups_ TAB_012	175	■0.995	<pre>\begin{tabular}[t]{ l l l l l } \hline \cos^2(\alpha, \beta) & \theta(\alpha, \beta) & n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha} & n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta} & \frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n} \\ \hline 1 & \frac{\pi}{2} \pm \frac{\pi}{2} & \pm 2 & \pm 2 & 1 \\ 3 & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 1 & \pm 3 & 3^{+1} \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 2 & \pm 1 & 2^{-1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 2 & 2^{+1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 1 & 1 \\ 0 & \frac{\pi}{2} & 0 & 0 & - \\ \hline \end{tabular}</pre>	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>3</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td></td><td></td><td>± 1</td><td>± 3</td><td>3^{+1}</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td></td><td></td><td>± 1</td><td>± 2</td><td>2^{+1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>-</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	3	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}			± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}			± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	-	
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																									
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3	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}																																									
		± 1	± 3	3^{+1}																																									
$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}																																									
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$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1																																									
0	$\frac{\pi}{2}$	0	0	-																																									

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																								
gilmore-lie-groups_ TAB_013	180	0.508	$\begin{tabular}{t}{ l l l l l l }\hline & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ \hline A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \\ \hline \end{tabular}$	<table><tr><td></td><td>α_1</td><td>α_2</td><td>α_3</td><td></td><td>α_{l-1}</td><td>α_l</td></tr><tr><td>A_{l-1}</td><td>$\mathbf{e}_1 - \mathbf{e}_2$</td><td>$\mathbf{e}_2 - \mathbf{e}_3$</td><td>$\mathbf{e}_3 - \mathbf{e}_4$</td><td>...</td><td>$\mathbf{e}_{l-1} - \mathbf{e}_l$</td><td></td></tr><tr><td>D_l</td><td>''</td><td>''</td><td>''</td><td>...</td><td>''</td><td>$\mathbf{e}_{l-1} + \mathbf{e}_l$</td></tr><tr><td>$B_l$</td><td>''</td><td>''</td><td>''</td><td>...</td><td>''</td><td>$\mathbf{e}_l$</td></tr><tr><td>C_l</td><td>''</td><td>''</td><td>''</td><td>...</td><td>''</td><td>$2\mathbf{e}_l$</td></tr></table>		α_1	α_2	α_3		α_{l-1}	α_l	A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	...	$\mathbf{e}_{l-1} - \mathbf{e}_l$		D_l	''	''	''	...	''	$\mathbf{e}_{l-1} + \mathbf{e}_l$	B_l	''	''	''	...	''	$\mathbf{e}_l$	C_l	''	''	''	...	''	$2\mathbf{e}_l$	$\begin{array}{ccccccc} & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & \mathbf{e}_l \\ C_l & \text{''} & \text{''} & \text{''} & \cdots & \text{''} & 2\mathbf{e}_l \end{array}$					
	α_1	α_2	α_3		α_{l-1}	α_l																																							
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C_l	''	''	''	...	''	$2\mathbf{e}_l$																																							
gilmore-lie-groups_ TAB_014	182	0.999	$\begin{tabular}{t}{ l l l l }\hline & (p) & (q) & (r) \\ \hline & \text{Root space} & & \\ \hline 3 & 3 & 2 & (E_{p+2}) \\ \hline 4 & 3 & 2 & (E_6) \\ \hline 5 & 3 & 2 & (E_7) \\ \hline 5 & 3 & 2 & (E_8) \\ \hline \end{tabular}$	<table><tr><td>p</td><td>q</td><td>r</td><td>Root space</td></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8	<table><tr><th>p</th><th>q</th><th>r</th><th>Root space</th></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8
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5	3	2	E_8																																										
gilmore-lie-groups_ TAB_015	184	1.000	$\begin{tabular}{t}{ l l l l }\hline & \text{Canonical form} & \text{Boson operators} & \text{Coordinates and derivatives} \\ \hline H_i & & b_i^\dagger b_i & x^i \partial_i \\ \hline E_{+\mathbf{e}_i - \mathbf{e}_j} & & b_i^\dagger b_j & x^i \partial_j \\ \hline \end{tabular}$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																
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gilmore-lie-groups_ TAB_016	184	■ 0.960	<pre>\begin{tabular}[t]{ l l l l l } \hline \multirow[b]{2}{*}{Canonical form} & \multicolumn{2}{ c }{\((C_l)\)} & \multicolumn{2}{ c }{\((D_l)\)} \\ \hline & \hline & Boson operators & Coordinates and derivatives & Fermion operators & Coordinates and derivatives \\ \hline H_i & $b_i^\dagger b_i + \frac{1}{2}$ & $x^i \partial_i$ & $f_i^\dagger f_i + \frac{1}{2}$ & $x^i \partial_i + \frac{1}{2}$ \\ $E_{+e_i - e_j}$ & $b_i^\dagger b_j$ & $x^i \partial_j$ & $f_i^\dagger f_j$ & $x^i \partial_j$ \\ $E_{+e_i + e_j}$ & $b_i^\dagger b_j^\dagger$ & $x^i x^j$ & $f_i^\dagger f_j^\dagger$ & $x^i x^j$ \\ $E_{-e_i - e_j}$ & $b_i b_j$ & $\partial_i \partial_j$ & $f_i f_j$ & $\partial_i \partial_j$ \\ E_{+2e_i} & $b_i^\dagger b_i^\dagger$ & $x^i x^i$ & & \\ E_{-2e_i} & $b_i b_i$ & $\partial_i \partial_i$ & & \\ \hline </pre>	(not rendered)	<table><tr><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td></tr><tr><td>$E_{+e_i - e_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td></tr><tr><td>$E_{+e_i + e_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td></tr><tr><td>$E_{-e_i - e_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td></tr><tr><td>$E_{+2e_i}$</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td></tr><tr><td>E_{-2e_i}</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td></tr></table>	C_l		D_l		Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives	H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$E_{+e_i - e_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$	$E_{+e_i + e_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$	$E_{-e_i - e_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$	E_{+2e_i}	$b_i^\dagger b_i^\dagger$	$x^i x^i$			E_{-2e_i}	$b_i b_i$	$\partial_i \partial_i$		
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gilmore-lie-groups_ TAB_017	195	■ 0.899	<pre>\begin{tabular}[t]{ l l l l l } \hline Mapping & Real form & Root space & Condition \\ \hline Block submatrix & $\mathfrak{so}(p, q)$ & D_n & $p + q = 2n$ \\ & $\mathfrak{so}(p, q)$ & B_n & $p + q = 2n + 1$ \\ & $\mathfrak{su}(p, q)$ & A_{n-1} & $p + q = n$ \\ & $\mathfrak{sp}(p, q)$ & C_n & $p + q = n$ \\ Subfield restriction & $\mathfrak{sl}(n; \mathbb{R})$ & A_{n-1} & \\ & $\mathfrak{sp}(2n; \mathbb{R})$ & C_n & \\ Field embedding & $\mathfrak{so}^*(2n)$ & D_n & \\ & $\mathfrak{su}^*(2n)$ & A_{2n-1} & \\ \hline </pre>	(not rendered)																																								

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gilmore-lie-groups_ TAB_018	195	■ 0.500	$\begin{tabular}{t l l l l l } \hline \scriptstyle(A_1) & \sim & \scriptstyle(B_1) & \sim & \scriptstyle(C_1) \\ \scriptstyle(u(2)) & \sim & \scriptstyle(so(3)) & \sim & \scriptstyle(sp(1) = usp(2)) \\ \scriptstyle(u(1,1) = sl(2; \mathbb{R})) & \sim & \scriptstyle(so(2,1)) & \sim & \scriptstyle(sp(2; \mathbb{R})) \\ \hline \scriptstyle(D_2) & = & \scriptstyle(A_1) & + & \scriptstyle(A_1) \\ \scriptstyle(so(4)) & = & \scriptstyle(so(3)) & + & \scriptstyle(so(3)) \\ \scriptstyle(so^*(4)) & \sim & \scriptstyle(so(3)) & + & \scriptstyle(so(2,1)) \\ \scriptstyle(so(3,1)) & \sim & \scriptstyle(sl(2; \mathbb{C})) & & \\ \scriptstyle(so(2,2)) & \sim & \scriptstyle(so(2,1)) & + & \scriptstyle(so(2,1)) \\ \hline \scriptstyle(B_2) & = & \scriptstyle(C_2) & & \\ \scriptstyle(so(5)) & \sim & \scriptstyle(sp(2) = usp(4)) & & \\ \scriptstyle(so(4,1)) & \sim & \scriptstyle(sp(1,1) = usp(2,2)) & & \\ \scriptstyle(so(3,2)) & \sim & \scriptstyle(sp(4; R)) & & \\ \hline \scriptstyle(D_3) & = & \scriptstyle(A_3) & & \\ \scriptstyle(so(6)) & \sim & \scriptstyle(su(4)) & & \\ \scriptstyle(so(5,1)) & \sim & \scriptstyle(su^*(4)) & & \\ \scriptstyle(so^*(6)) & \sim & \scriptstyle(su(3,1)) & & \\ \scriptstyle(so(4,2)) & \sim & \scriptstyle(su(2,2)) & & \\ \scriptstyle(so(3,3)) & \sim & \scriptstyle(sl(4; \mathbb{R})) & & \\ \hline \end{tabular}$	<table><tr><td>A_1</td><td></td><td>B_1</td><td></td><td>C_1</td></tr><tr><td>$su(2)$</td><td></td><td>$so(3)$</td><td></td><td>$sp(1) = usp(2)$</td></tr><tr><td>$su(1, 1) = sl(2; \mathbb{R})$</td><td></td><td>$so(2, 1)$</td><td></td><td>$sp(2; \mathbb{R})$</td></tr><tr><td>$D_2$</td><td>$=$</td><td>$A_1$</td><td>$+$</td><td>$A_1$</td></tr><tr><td>$so(4)$</td><td>$=$</td><td>$so(3)$</td><td>$+$</td><td>$so(3)$</td></tr><tr><td>$so^*(4)$</td><td></td><td>$so(3)$</td><td>$+$</td><td>$so(2, 1)$</td></tr><tr><td>$so(3, 1)$</td><td></td><td>$sl(2; \mathbb{C})$</td><td></td><td></td></tr><tr><td>$so(2, 2)$</td><td></td><td>$so(2, 1)$</td><td>$+$</td><td>$so(2, 1)$</td></tr><tr><td>B_2</td><td>$=$</td><td>C_2</td><td></td><td></td></tr><tr><td>$so(5)$</td><td></td><td>$sp(2) = usp(4)$</td><td></td><td></td></tr><tr><td>$so(4, 1)$</td><td></td><td>$sp(1, 1) = usp(2, 2)$</td><td></td><td></td></tr><tr><td>$so(3, 2)$</td><td></td><td>$sp(4; R)$</td><td></td><td></td></tr><tr><td>D_3</td><td>$=$</td><td>A_3</td><td></td><td></td></tr><tr><td>$so(6)$</td><td></td><td>$su(4)$</td><td></td><td></td></tr><tr><td>$so(5, 1)$</td><td></td><td>$su^*(4)$</td><td></td><td></td></tr><tr><td>$so^*(6)$</td><td></td><td>$su(3, 1)$</td><td></td><td></td></tr><tr><td>$so(4, 2)$</td><td></td><td>$su(2, 2)$</td><td></td><td></td></tr><tr><td>$so(3, 3)$</td><td></td><td>$sl(4; \mathbb{R})$</td><td></td><td></td></tr></table>	A_1		B_1		C_1	$su(2)$		$so(3)$		$sp(1) = usp(2)$	$su(1, 1) = sl(2; \mathbb{R})$		$so(2, 1)$		$sp(2; \mathbb{R})$	D_2	$=$	A_1	$+$	A_1	$so(4)$	$=$	$so(3)$	$+$	$so(3)$	$so^*(4)$		$so(3)$	$+$	$so(2, 1)$	$so(3, 1)$		$sl(2; \mathbb{C})$			$so(2, 2)$		$so(2, 1)$	$+$	$so(2, 1)$	B_2	$=$	C_2			$so(5)$		$sp(2) = usp(4)$			$so(4, 1)$		$sp(1, 1) = usp(2, 2)$			$so(3, 2)$		$sp(4; R)$			D_3	$=$	A_3			$so(6)$		$su(4)$			$so(5, 1)$		$su^*(4)$			$so^*(6)$		$su(3, 1)$			$so(4, 2)$		$su(2, 2)$			$so(3, 3)$		$sl(4; \mathbb{R})$			
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gilmore-lie-groups_ TAB_019	196	■1.000	<pre>\begin{tabular}[t]{ l l l l } \hline \multirow[b]{2}{*}{Root space} & \multirow[b]{2}{*}{Class \({}_\text{rank(character)}\)} & \multicolumn{2}{ l }{Maximal compact subgroup} \\ \hline & & Root space & Dimension \\ \hline G_2 & G_{2(-14)} & G_2 & 14 \\ & G_{2(+2)} & A_1 + A_1 & 6 \\ F_4 & F_{4(-52)} & F_4 & 52 \\ & F_{4(-20)} & B_4 & 36 \\ & F_{4(+4)} & C_3 + A_1 & 24 \\ E_6 & E_{6(-78)} & E_6 & 78 \\ & E_{6(-26)} & F_4 & 52 \\ & E_{6(-14)} & D_5 + D_1 & 46 \\ & E_{6(+2)} & A_5 + A_1 & 38 \\ & E_{6(+6)} & C_4 & 36 \\ E_7 & E_{7(-133)} & E_7 & 133 \\ & E_{7(-25)} & E_6 + D_1 & 79 \\ & E_{7(-5)} & D_6 + A_1 & 69 \\ & E_{7(+7)} & A_7 & 63 \\ E_8 & E_{8(-248)} & E_8 & 248 \\ & E_{8(-24)} & E_7 + A_1 & 136 \\ & E_{8(+8)} & D_8 & 120 \\ \hline \hline \end{tabular}</pre>	(not rendered)	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$															
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gilmore-lie-groups_ TAB_020	199	■1.000	<pre>\begin{tabular}[t]{ l l l l } \hline Quadrant & Operator & Edge \\ \hline I & \({}_J\{+\}\) & \({}_n\{2\}=0\} \\ \hline I & \({}_J\{-\}\) & \({}_n\{1\}=0\} \\ \hline III & \({}_J\{+\}\) & \({}_n\{1\}=-1\} \\ \hline III & \({}_J\{-\}\) & \({}_n\{2\}=-1\} \\ \hline \end{tabular}</pre>	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$
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gilmore-lie-groups_021	206	0.877	$\begin{tabular}{t l l l l l } \hline \text{Root space} \& \text{Quotient } (G^{\prime}/H) \& \text{Dimension } (P) \& \text{Rank } (P) \\ \hline (A_{p+q-1}) \& (SU(p, q)/S[U(p) \otimes U(q)]) \& (2pq) \& \min(p, q) \\ (A_{n-1}) \& (SL(n; \mathbb{R})/SO(n)) \& (\frac{1}{2}(n+2)(n-1)) \& n-1 \\ (A_{2n-1}) \& (SU^*(2n)/U\operatorname{Sp}(2n)) \& ((2n+1)(n-1)) \& n-1 \\ (B_{p+q}) \& (SO(p, q)/SO(p) \otimes SO(q)) \& (pq) \& \min(p, q) \\ (D_{p+q}) \& (SO(p, q)/SO(p) \otimes SO(q)) \& (pq) \& \min(p, q) \\ (D_n) \& (SO^*(2n)/U(n)) \& (n(n-1)) \& [n/2] \\ (C_{p+q}) \& (U\operatorname{Sp}(2p, 2q)/U\operatorname{Sp}(2p) \otimes U\operatorname{Sp}(2q)) \& (4pq) \& \min(p, q) \\ (C_n) \& (Sp(2n; \mathbb{R})/U(n)) \& (n(n+1)) \& n \\ \hline \end{tabular}$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/U\operatorname{Sp}(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\operatorname{Sp}(2p, 2q)/U\operatorname{Sp}(2p) \otimes U\operatorname{Sp}(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\operatorname{Sp}(2p, 2q)/U\operatorname{Sp}(2p) \otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/U\operatorname{Sp}(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\operatorname{Sp}(2p, 2q)/U\operatorname{Sp}(2p) \otimes U\operatorname{Sp}(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/U\operatorname{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\operatorname{Sp}(2p, 2q)/U\operatorname{Sp}(2p) \otimes U\operatorname{Sp}(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n						
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gilmore-lie-groups_022	207	0.979	$\begin{tabular}{t l l l l l } \hline \text{Root space} \& (G^{\prime}/H) \& (\operatorname{Dim} G^{\prime}) \& (\operatorname{Dim} H) \& (\operatorname{Dim} P) \& \text{Rank } (P) \\ \hline (G_2) \& (G_{2(+2)}) \& (A_1 \oplus A_1) \& 14 \& 6 \& 8 \& 2 \\ (F_4) \& (F_{4(-20)}) \& B_4 \& 52 \& 36 \& 16 \& 1 \\ (F_4) \& (F_{4(+4)}) \& (C_3 \oplus A_1) \& 52 \& 24 \& 28 \& 4 \\ (E_6) \& (E_{6(-26)}) \& F_4 \& 78 \& 52 \& 26 \& 2 \\ (E_6) \& (E_{6(-14)}) \& (D_5 \oplus D_1) \& 78 \& 46 \& 32 \& 2 \\ (E_6) \& (E_{6(+2)}) \& (A_5 \oplus A_1) \& 78 \& 38 \& 40 \& 4 \\ (E_6) \& (E_{6(+6)}) \& C_4 \& 78 \& 36 \& 42 \& 6 \\ (E_7) \& (E_{7(-25)}) \& (E_6 \oplus D_1) \& 133 \& 79 \& 54 \& 3 \\ (E_7) \& (E_{7(-5)}) \& (D_6 \oplus A_1) \& 133 \& 69 \& 64 \& 4 \\ (E_7) \& (E_{7(+7)}) \& A_7 \& 133 \& 63 \& 70 \& 7 \\ (E_8) \& (E_{8(-24)}) \& (E_7 \oplus A_1) \& 248 \& 136 \& 112 \& 4 \\ (E_8) \& (E_{8(+8)}) \& D_8 \& 248 \& 120 \& 128 \& 8 \\ \hline \end{tabular}$	(not rendered)	<table><tr><th>Root space</th><th>G'/H</th><th>$\operatorname{Dim} G'$</th><th>$\operatorname{Dim} H$</th><th>$\operatorname{Dim} P$</th><th>$\operatorname{Rank} P$</th></tr><tr><td>G_2</td><td>$G_{2(+2)}/(A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr><tr><td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr><tr><td></td><td>$F_{4(+4)}/(C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr><tr><td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr><tr><td></td><td>$E_{6(-14)}/(D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr><tr><td></td><td>$E_{6(+2)}/(A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr><tr><td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr><tr><td>E_7</td><td>$E_{7(-25)}/(E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr><tr><td></td><td>$E_{7(-5)}/(D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr><tr><td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr><tr><td>E_8</td><td>$E_{8(-24)}/(E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr><tr><td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr></table>	Root space	G'/H	$\operatorname{Dim} G'$	$\operatorname{Dim} H$	$\operatorname{Dim} P$	$\operatorname{Rank} P$	G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																																
gilmore-lie-groups_ TAB_023	212	■ 1.000	$\begin{tabular}{t}{ l l l l } \hline \text{Operation} \& \backslash(i \sigma_1 \backslash) \& \backslash(i \sigma_2 \backslash) \& \backslash(i \sigma_3 \backslash) \\ \hline \backslash(T_1=\backslash) \text{block matrix decomposition} \& -1 \& -1 \& +1 \\ \hline \backslash(T_2=\backslash) \text{complex conjugation} \& -1 \& +1 \& -1 \\ \hline \backslash(T_3=T_1 T_2) \& +1 \& -1 \& -1 \\ \hline \end{tabular}$	<table><tr><th>Operation</th><th>$i\sigma_1$</th><th>$i\sigma_2$</th><th>$i\sigma_3$</th></tr><tr><td>$T_1 = \text{block matrix decomposition}$</td><td>-1</td><td>-1</td><td>+1</td></tr><tr><td>$T_2 = \text{complex conjugation}$</td><td>-1</td><td>+1</td><td>-1</td></tr><tr><td>$T_3 = T_1 T_2$</td><td>+1</td><td>-1</td><td>-1</td></tr></table>	Operation	$i\sigma_1$	$i\sigma_2$	$i\sigma_3$	$T_1 = \text{block matrix decomposition}$	-1	-1	+1	$T_2 = \text{complex conjugation}$	-1	+1	-1	$T_3 = T_1 T_2$	+1	-1	-1	<table><tr><th>Operation</th><th>$i\sigma_1$</th><th>$i\sigma_2$</th><th>$i\sigma_3$</th></tr><tr><td>$T_1 = \text{block matrix decomposition}$</td><td>-1</td><td>-1</td><td>+1</td></tr><tr><td>$T_2 = \text{complex conjugation}$</td><td>-1</td><td>+1</td><td>-1</td></tr><tr><td>$T_3 = T_1 T_2$</td><td>+1</td><td>-1</td><td>-1</td></tr></table>	Operation	$i\sigma_1$	$i\sigma_2$	$i\sigma_3$	$T_1 = \text{block matrix decomposition}$	-1	-1	+1	$T_2 = \text{complex conjugation}$	-1	+1	-1	$T_3 = T_1 T_2$	+1	-1	-1
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gilmore-lie-groups_ TAB_024	232	■ 1.000	$\begin{tabular}{t}{ l l l l } \hline \text{Bessel function} \& \rightarrow \& \text{harmonic oscillator eigenfunction} \\ \hline \text{Bessel equation} \& \rightarrow \& \text{Schrödinger equation for harmonic oscillator} \\ \hline \end{tabular}$	<table><tr><td>Bessel function</td><td>→</td><td>harmonic oscillator eigenfunction</td></tr><tr><td>Bessel equation</td><td>→</td><td>Schrödinger equation for harmonic oscillator</td></tr></table>	Bessel function	→	harmonic oscillator eigenfunction	Bessel equation	→	Schrödinger equation for harmonic oscillator	<table><tr><td>Bessel function</td><td>→</td><td>harmonic oscillator eigenfunction</td></tr><tr><td>Bessel equation</td><td>→</td><td>Schrödinger equation for harmonic oscillator</td></tr></table>	Bessel function	→	harmonic oscillator eigenfunction	Bessel equation	→	Schrödinger equation for harmonic oscillator																				
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gilmore-lie-groups_ TAB_025	239	■ 0.424	$\begin{tabular}{t}{ l l l l l } \hline \text{Equation} \& \backslash(A \backslash) \& \backslash(B \backslash) \& \backslash(C \backslash) \\ \hline \text{Klein-Gordon} \& \backslash(-l(l+1)+\left(e^2 / \hbar c\right)^2 \backslash) \& \backslash(2 E e^2 /(\hbar c) \backslash) \& \backslash(\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2 \backslash) \\ \hline \text{Schrödinger} \& \backslash(-l(l+1) \backslash) \& \backslash(2 m e^2 / \hbar^2 \backslash) \& \backslash(2 m \mathrm{~} W / \hbar^2 \backslash) \\ \hline \end{tabular}$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein-Gordon</td><td>$-l(l+1)+(e^2/\hbar c)^2$</td><td>$2Ee^2/(\hbar c)^2$</td><td>$[E^2-(mc^2)^2]/(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2me^2/\hbar^2$</td><td>$2mW/\hbar^2$</td></tr></table>	Equation	A	B	C	Klein-Gordon	$-l(l+1)+(e^2/\hbar c)^2$	$2Ee^2/(\hbar c)^2$	$[E^2-(mc^2)^2]/(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein–Gordon</td><td>$-l(l+1)+(e^2/\hbar c)^2$</td><td>$2Ee^2/(\hbar c)^2$</td><td>$[E^2-(mc^2)^2]/(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2me^2/\hbar^2$</td><td>$2mW/\hbar^2$</td></tr></table>	Equation	A	B	C	Klein–Gordon	$-l(l+1)+(e^2/\hbar c)^2$	$2Ee^2/(\hbar c)^2$	$[E^2-(mc^2)^2]/(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$								
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gilmore-lie-groups_ TAB_026	240	■ 0.655	\begin{tabular}[t]{ l l l } \hline & Procedure & Result \\ \hline 1 & Locate singularities & $0, \infty$ \\ 2 & Determine analytic behavior at singular points & $r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$ \\ 3 & Keep only $\mathcal{L}^2$ solutions & $\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$ \\ 4 & Look for solutions with proper asymptotic behavior & $R = r^\gamma e^{\lambda r} f(r)$ \\ 5 & Construct differential equation for $f(r)$ & $[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$ \\ 6 & Construct recursion relation & $f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$ \\ 7 & Look at asymptotic behavior & $f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$) \\ 8 & Construct quantization condition & $2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$ \\ 9 & Construct explicit solutions & $E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$ \\ \hline \end{tabular}	(not rendered)	<table><tr><th>Procedure</th><th>Result</th></tr><tr><td>1 Locate singularities</td><td>$0, \infty$</td></tr><tr><td>2 Determine analytic behavior at singular points</td><td>$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$</td></tr><tr><td>3 Keep only $\mathcal{L}^2$ solutions</td><td>$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$</td></tr><tr><td>4 Look for solutions with proper asymptotic behavior</td><td>$R = r^\gamma e^{\lambda r} f(r)$</td></tr><tr><td>5 Construct differential equation for $f(r)$</td><td>$[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$</td></tr><tr><td>6 Construct recursion relation</td><td>$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$</td></tr><tr><td>7 Look at asymptotic behavior</td><td>$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)</td></tr><tr><td>8 Construct quantization condition</td><td>$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$</td></tr><tr><td>9 Construct explicit solutions</td><td>$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$</td></tr></table>	Procedure	Result	1 Locate singularities	$0, \infty$	2 Determine analytic behavior at singular points	$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$	3 Keep only $\mathcal{L}^2$ solutions	$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$	4 Look for solutions with proper asymptotic behavior	$R = r^\gamma e^{\lambda r} f(r)$	5 Construct differential equation for $f(r)$	$[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$	6 Construct recursion relation	$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$	7 Look at asymptotic behavior	$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)	8 Construct quantization condition	$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$	9 Construct explicit solutions	$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$
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gilmore-lie-groups_TAB_027	244	■0.998	$\begin{tabular}{t l l l l } \hline \text{\(m\)} & \text{\(l=0\)} & \text{\(l=1\)} & \text{\(l=2\)} & \text{\(l=3\)} \\ \hline 0 & \sqrt{\frac{1}{4\pi}} & \sqrt{\frac{3}{4\pi}}\cos\theta & \sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1) & \sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta) \\ \pm 1 & & \mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi} & \mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi} & \mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi} \\ \pm 2 & & & \sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi} & \sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi} \\ \pm 3 & & & & \mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi} \\ \hline \hline \end{tabular}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$				
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gilmore-lie-groups_TAB_028	258	■0.721	$\begin{tabular}{t l l l } \hline H & \text{\(A_4\)} & \text{\(\Gamma_4\)} & \text{\(\Gamma_5\)} \\ \hline A & \text{\(\Theta\)} & \text{\(A\)} & \text{\(B\)} \\ \hline B & \text{\(A\)} & \text{\(B\)} & \text{\(C\)} \\ \hline C & \text{\(A\)} & \text{\(B\)} & \text{\(C\)} \\ \hline \hline \end{tabular}$	<table><tr><th>H</th><th>$\frac{p^2}{2m} - \frac{\alpha}{r}$</th><th>$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$</th></tr><tr><td>$A_4$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$</td><td>$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$</td><td>$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Theta$</td><td>$r(H_S - W)$</td><td>$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$</td></tr><tr><td></td><td>$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$</td><td>$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$</td></tr><tr><td>$A$</td><td>$1/2m$</td><td>$1$</td></tr><tr><td>$B$</td><td>$-W$</td><td>$m^2 - E^2$</td></tr><tr><td>$C$</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m} - \frac{\alpha}{r}$	$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$	A_4	$\mathbf{r} \cdot \mathbf{p} - i$	$\mathbf{r} \cdot \mathbf{p} - i$	Γ_4	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$	Γ_5	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$	Θ	$r(H_S - W)$	$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$		$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	A	$1/2m$	1	B	$-W$	$m^2 - E^2$	C	$-\alpha$	$-2\alpha E$	<table><tr><th>H</th><th>$\frac{p^2}{2m} - \frac{\alpha}{r}$</th><th>$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$</th></tr><tr><td>$A_4$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$</td><td>$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$</td><td>$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Theta$</td><td>$r(H_S - W)$</td><td>$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$</td></tr><tr><td></td><td>$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$</td><td>$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$</td></tr><tr><td>$A$</td><td>$1/2m$</td><td>$1$</td></tr><tr><td>$B$</td><td>$-W$</td><td>$m^2 - E^2$</td></tr><tr><td>$C$</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m} - \frac{\alpha}{r}$	$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$	A_4	$\mathbf{r} \cdot \mathbf{p} - i$	$\mathbf{r} \cdot \mathbf{p} - i$	Γ_4	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$	Γ_5	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$	Θ	$r(H_S - W)$	$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$		$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	A	$1/2m$	1	B	$-W$	$m^2 - E^2$	C	$-\alpha$	$-2\alpha E$
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gilmore-lie-groups_ TAB_029	267	■0.953	$\begin{tabular}[t]{ l l }\hline Particle & Rest energy (MeV) \\ \hline electron e^\pm & 0.511 \\ \hline mu meson \mu^\pm & 105.7 \\ \hline tau meson \tau^\pm & 1784.0 \\ \hline proton, antiproton p^\pm & 938.26 \\ \hline deuteron d^+ & 1875.6 \\ \hline tritium t^+ & 2809.4 \\ \hline pi meson \pi^\pm & 139.6 \\ \hline sigma meson \Sigma^\pm & 1385.0 \\ \hline cascade meson \Xi^- & 1533.0 \\ \hline omega \Omega^- & 1672.0 \\ \hline\end{tabular}$	<table><tr><th>Particle</th><th>Rest energy (MeV)</th></tr><tr><td>electron $e^\pm$</td><td>0.511</td></tr><tr><td>mu meson $\mu^\pm$</td><td>105.7</td></tr><tr><td>tau meson $\tau^\pm$</td><td>1784.0</td></tr><tr><td>proton, antiproton $p^\pm$</td><td>938.26</td></tr><tr><td>deuteron d^+</td><td>1875.6</td></tr><tr><td>tritium t^+</td><td>2809.4</td></tr><tr><td>pi meson $\pi^\pm$</td><td>139.6</td></tr><tr><td>sigma meson $\Sigma^\pm$</td><td>1385.0</td></tr><tr><td>cascade meson Ξ^-</td><td>1533.0</td></tr><tr><td>omega Ω^-</td><td>1672.0</td></tr></table>	Particle	Rest energy (MeV)	electron $e^\pm$	0.511	mu meson $\mu^\pm$	105.7	tau meson $\tau^\pm$	1784.0	proton, antiproton $p^\pm$	938.26	deuteron d^+	1875.6	tritium t^+	2809.4	pi meson $\pi^\pm$	139.6	sigma meson $\Sigma^\pm$	1385.0	cascade meson Ξ^-	1533.0	omega Ω^-	1672.0	<table><tr><th>Particle</th><th>Rest energy (MeV)</th></tr><tr><td>electron $e^\pm$</td><td>0.511</td></tr><tr><td>mu meson $\mu^\pm$</td><td>105.7</td></tr><tr><td>tau meson $\tau^\pm$</td><td>1784.0</td></tr><tr><td>proton, antiproton $p^\pm$</td><td>938.26</td></tr><tr><td>deuteron d^+</td><td>1875.6</td></tr><tr><td>tritium t^+</td><td>2809.4</td></tr><tr><td>pi meson $\pi^\pm$</td><td>139.6</td></tr><tr><td>sigma meson $\Sigma^\pm$</td><td>1385.0</td></tr><tr><td>cascade meson Ξ^-</td><td>1533.0</td></tr><tr><td>omega Ω^-</td><td>1672.0</td></tr></table>	Particle	Rest energy (MeV)	electron $e^\pm$	0.511	mu meson $\mu^\pm$	105.7	tau meson $\tau^\pm$	1784.0	proton, antiproton $p^\pm$	938.26	deuteron d^+	1875.6	tritium t^+	2809.4	pi meson $\pi^\pm$	139.6	sigma meson $\Sigma^\pm$	1385.0	cascade meson Ξ^-	1533.0	omega Ω^-	1672.0
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gilmore-lie-groups_ TAB_030	269	■1.000	$\begin{tabular}[t]{ l l l l l l l l }\hline Mercury & Venus & Earth & Mars & Jupiter & Saturn & Uranus & Neptune \\ \hline 0.39 & 0.72 & 1.00 & 1.52 & 5.20 & 9.54 & 19.18 & 30.06 \\ \hline\end{tabular}$	<table><tr><th>Mercury</th><th>Venus</th><th>Earth</th><th>Mars</th><th>Jupiter</th><th>Saturn</th><th>Uranus</th><th>Neptune</th></tr><tr><td>0.39</td><td>0.72</td><td>1.00</td><td>1.52</td><td>5.20</td><td>9.54</td><td>19.18</td><td>30.06</td></tr></table>	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune	0.39	0.72	1.00	1.52	5.20	9.54	19.18	30.06	<table><tr><th>Mercury</th><th>Venus</th><th>Earth</th><th>Mars</th><th>Jupiter</th><th>Saturn</th><th>Uranus</th><th>Neptune</th></tr><tr><td>0.39</td><td>0.72</td><td>1.00</td><td>1.52</td><td>5.20</td><td>9.54</td><td>19.18</td><td>30.06</td></tr></table>	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune	0.39	0.72	1.00	1.52	5.20	9.54	19.18	30.06												
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gilmore-lie-groups_ TAB_032	280	■0.815	$\begin{tabular}[t]{ l l l }\hline & k\rangle & \left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle \\ \hline \{I, a\} & k\rangle e^{ik\cdot a} & \left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'} \\ \hline \{\Lambda, 0\} & \Lambda k\rangle & \left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D^{jj'}_{vv';\mu\mu'}(\Lambda) \\ \hline\end{tabular}$	<table><tr><th></th><th>$k\rangle$</th><th>$\left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle$</th></tr><tr><td>$\{I, a\}$</td><td>$k\rangle e^{ik\cdot a}$</td><td>$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'}$</td></tr><tr><td>$\{\Lambda, 0\}$</td><td>$\Lambda k\rangle$</td><td>$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D^{jj'}_{vv';\mu\mu'}(\Lambda)$</td></tr></table>		$ k\rangle$	$\left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle$	$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'}$	$\{\Lambda, 0\}$	$ \Lambda k\rangle$	$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D^{jj'}_{vv';\mu\mu'}(\Lambda)$	<table><tr><th></th><th>$k\rangle$</th><th>$\left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle$</th></tr><tr><td>$\{I, a\}$</td><td>$k\rangle e^{ik\cdot a}$</td><td>$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'}$</td></tr><tr><td>$\{\Lambda, 0\}$</td><td>$\Lambda k\rangle$</td><td>$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D^{jj'}_{vv';\mu\mu'}(\Lambda)$</td></tr></table>		$ k\rangle$	$\left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle$	$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'}$	$\{\Lambda, 0\}$	$ \Lambda k\rangle$	$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D^{jj'}_{vv';\mu\mu'}(\Lambda)$																										
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																								
gilmore-lie-groups_ TAB_033	290	■ 1.000	<pre>\begin{tabular}[t]{ l l l } \hline & Electric & Magnetic \\ \hline Charge density & \(\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)\) & \(\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)\) \\ Current density & \(\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}\) & \(\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}\) \\ Conservation law & \(\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0\) & \(\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0\) \\ \hline \end{tabular}</pre>	<table><tr><th></th><th>Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$	<table><tr><th colspan="2">Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>	Electric		Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$
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gilmore-lie-groups_TAB_034	311	0.852	$\begin{tabular}{ t { l l l l l l }} \hline \multicolumn{3}{ c }{\text{Infinitesimal generators}} \& \multicolumn{3}{ c }{\text{Canonical coordinates}} \& \text{Constraint} \\ \hline \xi(x,y) \& \eta(x,y) \& \zeta(x,y,p) \& R(x,y) \& S(x,y) \& T(x,y,p) \& dS/dR \\ \hline 0 \& 1 \& 0 \& x \& y \& p \& T \\ 1 \& 0 \& 0 \& y \& x \& p \& 1/T \\ 1/a \& -1/b \& 0 \& ax+by \& bx-ay \& p \& (b-aT)/(a+bT) \\ x \& 0 \& -p \& y \& \ln x \& xp \& 1/T \\ 0 \& y \& p \& x \& \ln y \& p/y \& T \\ x/a \& y/b \& (1/b-1/a)p \& y^b/x^a \& b \ln y \& p/x^{(a/b-1)} \& (bT/R)/(bT-aR^{(1/b)}) \\ x \& y \& 0 \& y/x \& \ln y \& p \& (T/R)/(T-R) \\ x \& -y \& -2p \& xy \& \ln y \& x^2p \& (T/R)/(T+R) \\ 2x \& y \& -p \& y^2/x \& \ln y \& yp \& (T/R)/(2T-R) \\ x \& 2y \& p \& y/x^2 \& \ln y \& p/x \& (T/R)/(T-2R) \\ y \& 0 \& -p^2 \& y \& x/y \& x-y/p \& -T/R^2 \\ 0 \& x \& 1 \& x \& y/x \& xp-y \& T/R^2 \\ -y \& x \& 1+p^2 \& \sqrt{x^2+y^2} \& \tan^{-1}(y/x) \& (y-xp)/(x+yp) \& -T/R \\ 1 \& y/x \& (px-y)/x^2 \& y/x \& x \& (xp-y)/x^2 \& 1/T \\ a \& x \& 1 \& x^2-2ay \& x/a \& x-ap \& 1/(2aT) \\ a \& y \& p \& x-a \ln y \& x/a \& p/y \& (1/a)/(1-aT) \\ x \& b \& -p \& e^y/x^b \& y/b \& e^y * p^b \& (bR)^{-1}/[1-b(R/T)^{(1/b)}] \\ \\ y \& b \& -p^2 \& y^2-2bx \& y/b \& y-b/p \& 1/(2bT) \\ 0 \& e^{f(x)} \& f'e^f \& x \& y/e^f \& p-yf' \& T/e^{f(R)} \\ x^2 \& xy \& y-xp \& y/x \& 1/x \& xp-y \& 1/T \\ xy \& y^2 \& yp-xp^2 \& y/x \& 1/y \& y/p-x \& 1/(TR^2) \\ xy \& 0 \& -yp-xp^2 \& y \& (\ln x)/y \& y/(xp)-\ln x \& T/R^2 \\ 0 \& xy \& y+xp \& x \& (\ln y)/x \& xp/y-\ln y \& T/R^2 \\ g(y) \& 0 \& -g'p^2 \& y \& x/g \& 1/p-xg'/g \& T/g(R) \\ 0 \& f(x) \& f' \& x \& y/f \& fp-f'y \& T/f^2(R) \\ f(x) \& 0 \& -f'p \& y \& F \text{ (} F'f=1 \text{)} \& pf \& 1/T \\ 0 \& g(y) \& g'p \& x \& G \text{ (} G'g=1 \text{)} \& p/g \& T \\ x^{k+1} \& kx^ky \& x^k(k^2y/x-p) \& y/x^k \& 1/x^k \& xp-ky \& -k/T \\ kxy^k \& y^{k+1} \& y^k(p-k^2xp^2/y) \& x/y^k \& 1/y^k \& y/p-kx \& -k/T \\ \hline \end{tabular}$	<table><tr><th colspan="2">Infinitesimal generators</th><th colspan="2">Canonical coordinates</th><th>Constraint</th></tr><tr><td>$\xi(x,y)$</td><td>$\eta(x,y)$</td><td>$R(x,y)$</td><td>$S(x,y)$</td><td>$T(x,y,p)$</td></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a \ln y$</td><td>x/a</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td></tr><tr><td>y</td><td>b</td><td>$-p^2$</td><td>y^2-2bx</td><td>y/b</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln 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y$	y	0	$-p^2$	y	x/y	0	x	1	x	y/x	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	1	y/x	$(px-y)/x^2$	y/x	x	a	x	1	x^2-2ay	x/a	a	y	p	$x-a \ln y$	x/a	x	b	$-p$	e^y/x^b	y/b	y	b	$-p^2$	y^2-2bx	y/b	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	x^2	xy	$y-xp$	y/x	$1/x$	xy	y^2	$yp-xp^2$	y/x	$1/y$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	0	xy	$y+xp$	x	$(\ln y)/x$	$g(y)$	0	$-g'p^2$	y	x/g	0	$f(x)$	f'	x	y/f	$f(x)$	0	$-f'p$	y	$F \text{ (} F'f=1 \text{)}$	0	$g(y)$	$g'p$	x	$G \text{ (} G'g=1 \text{)}$	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$
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kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$																																																																																																																																																											

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gilmore-lie-groups_ TAB_035	316	■1.000	$\begin{tabular}{t}{ l l l }\hline 1830 & \text{Galois} & \text{solve algebraic equations} \\ \hline 1874 & \text{Lie} & \text{solve differential equations} \\ \hline 1969 & \text{Risch} & \text{integrate in closed form} \\ \hline ? & - & \text{solve determining equations for } \xi, \eta \\ \hline ? & - & \text{solve determining equations for } R, S, T \\ \hline ? & - & \text{solve } F(R, -, T) = 0 \text{ for } R. \\ \hline \end{tabular}$	<table><tr><td>1830</td><td>Galois</td><td>solve algebraic equations</td></tr><tr><td>1874</td><td>Lie</td><td>solve differential equations</td></tr><tr><td>1969</td><td>Risch</td><td>integrate in closed form</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for ξ, η</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for R, S, T</td></tr><tr><td>?</td><td>-</td><td>solve $F(R, -, T) = 0$ for R.</td></tr></table>	1830	Galois	solve algebraic equations	1874	Lie	solve differential equations	1969	Risch	integrate in closed form	?	-	solve determining equations for ξ, η	?	-	solve determining equations for R, S, T	?	-	solve $F(R, -, T) = 0$ for R .	<table><tr><td>1830</td><td>Galois</td><td>solve algebraic equations</td></tr><tr><td>1874</td><td>Lie</td><td>solve differential equations</td></tr><tr><td>1969</td><td>Risch</td><td>integrate in closed form</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for ξ, η</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for R, S, T</td></tr><tr><td>?</td><td>-</td><td>solve $F(R, -, T) = 0$ for R.</td></tr></table>	1830	Galois	solve algebraic equations	1874	Lie	solve differential equations	1969	Risch	integrate in closed form	?	-	solve determining equations for ξ, η	?	-	solve determining equations for R, S, T	?	-	solve $F(R, -, T) = 0$ for R .						
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gilmore-lie-groups_ TAB_036	318	■1.000	$\begin{tabular}{t}{ l l l }\hline \text{\(k\)} & \text{Potential type} & \text{Transformation} \\ \hline -1 & \text{Coulomb} & \gamma^2 = \beta^3 \\ \hline 0 & \text{no force} & \gamma^2 = \beta^2 \\ \hline +1 & \text{local gravitational potential} & \gamma^2 = \beta^1 \\ \hline +2 & \text{harmonic oscillator} & \gamma^2 = \beta^0 \\ \hline \end{tabular}$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$												
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gilmore-lie-groups_ TAB_038	320	■1.000	$\begin{tabular}{t}{ l l l }\hline \text{Symmetry} & \text{Conserved quantity} \\ \hline \text{Space displacements} & \text{momentum} \\ \hline \text{Time displacements} & \text{energy} \\ \hline \text{Space-time displacements} & \text{four-momentum} \\ \hline \text{Rotations} & \text{angular momentum} \\ \hline \end{tabular}$	<table><tr><th>Symmetry</th><th>Conserved quantity</th></tr><tr><td>Space displacements</td><td>momentum</td></tr><tr><td>Time displacements</td><td>energy</td></tr><tr><td>Space-time displacements</td><td>four-momentum</td></tr><tr><td>Rotations</td><td>angular momentum</td></tr></table>	Symmetry	Conserved quantity	Space displacements	momentum	Time displacements	energy	Space-time displacements	four-momentum	Rotations	angular momentum	<table><tr><th>Symmetry</th><th>Conserved quantity</th></tr><tr><td>Space displacements</td><td>momentum</td></tr><tr><td>Time displacements</td><td>energy</td></tr><tr><td>Space-time displacements</td><td>four-momentum</td></tr><tr><td>Rotations</td><td>angular momentum</td></tr></table>	Symmetry	Conserved quantity	Space displacements	momentum	Time displacements	energy	Space-time displacements	four-momentum	Rotations	angular momentum																						
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